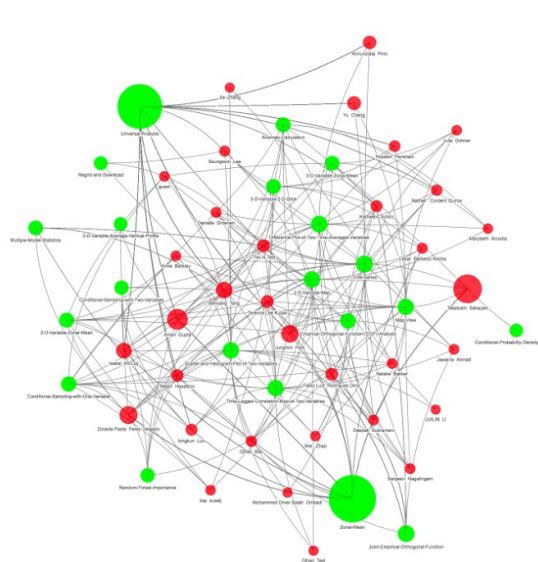
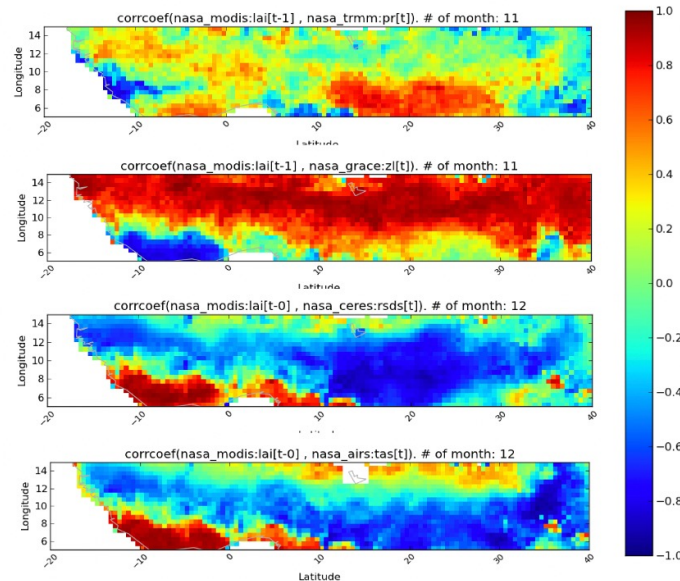
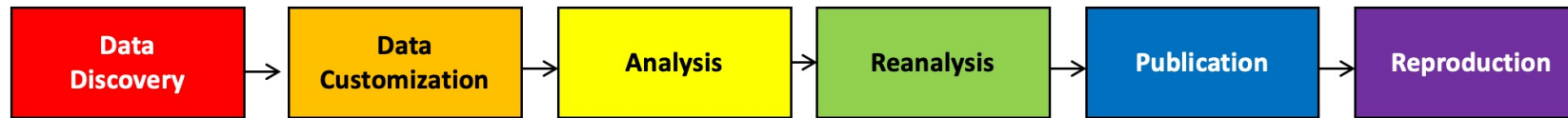


Student Group Projects for 2023 NASA Summer School on Satellite Observations and Climate Models

Lead: Seungwon Lee

Engineer: Alex Goodman

Scientists: Ou Wang, Hanii Takahashi, Jonathan Jiang, JT Reager, Kay Suselj, Kyo Lee, Ben Hamlington



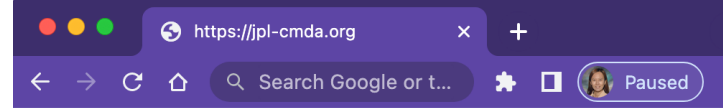
Student Group Projects

- Hands-on experience with climate science topics.
- Seven group research topics have been designed.
- Project information can be found in the project server: <https://jpl-cmda.org>.
- All the datasets and analysis tools needed are provided in the project server: <https://jpl-cmda.org>.
- The server has two interfaces: (1) Web HTML interface and (2) Jupyter Notebook interface.
- One introduction session now (datasets, tools, topics, group formation).
- Four practice sessions on 8/17 (Today), 8/21 (Mon), 8/22 (Tues), 8/24 (Thurs).
- Final presentation session on 8/24 (Thurs).



Group Project Server

- Server: <https://jpl-cmda.org/>
- Two interfaces:
 - Webservice HTML interface
 - Jupyter Notebook interface
- Jupyter Server Login information
 - Username: your email address
 - Password: nasa-ccs
- Jupyter Server Selection
 - CMDA: choose for Topics 1,2,3,4,5
 - RCMES: choose for Topic 6



Climate Model Diagnostic Analyzer (CMDA) is a collaborative platform to support the full life cycle of a data analysis process, from data discovery, to data customization, to analysis, to reanalysis, to publication/sharing, and to reproduction [1,2]. CMDA was initially developed to demonstrate the methodology to evaluate and diagnose climate models through the comprehensive use of multiple observational data, reanalysis data, and model outputs. It has evolved to support collaborative scientific activities and the full life cycle of data analysis. The CMDA project has been funded by the NASA ROSES programs.

CMDA HTML Interface

The CMDA HTML interface provides a user with a full control on input parameter selections for a given service such as dataset selection, dataset subsetting condition, dataset analysis functionality, and output data visualization parameters.

CMDA Jupyter-Notebook Interface

The CMDA Jupyter Notebook Interface allows users to make direct API requests to the CMDA webservices to retrieve the datasets of their choice and perform their own analysis on a Jupyter Hub server.

CMDA Datasets

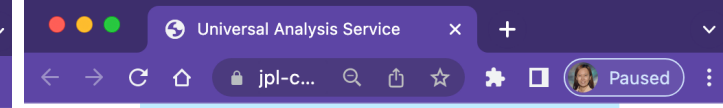
CMDA hosts over 2000 datasets covering model datasets, observational datasets, and reanalysis datasets. The model datasets include CMIP5 historical runs, CMIP5 AMIP runs, CMIP5 RCP4.5 projection runs, and WRF model runs with various physical parameterizations. The observation datasets cover many satellite data (AIRS, AMSR-E, AVISO, CERES, GRACE, GPCP, GPM, ISCCP, MISR, MODIS, MLS, QuickSCAT, SMAP, TES, TRMM) and ship-floats data (ARGO). The reanalysis datasets include ECMWF and GLDAP data.

CMDA Application: NASA Summer School Group Projects

CMDA has been successfully utilized in a real-world user environment. Since 2014, the annual NASA Summer School on Satellite Observations and Climate Models has used CMDA as a collaborative computing platform for students to conduct their group projects. The summer school students were given a short introduction to CMDA and were able to use it to access datasets and to analyze the datasets to generate their group project result within one week of the summer school. CMDA leveraged Amazon Elastic Kubernetes Service clusters to scale up its computation resources according to the demands from the students.

References

- [1] [Educational and Scientific Applications of Climate Model Diagnostic Analyzer, Seungwon Lee et al., IEEE International Congress on Big Data, Honolulu, HI, June, 2017.](#)
- [2] [Climate Model Diagnostic Analyzer, Seungwon Lee et al., IEEE International Conference on Big Data, Santa Clara, CA, October, 2015.](#)



Universal Analysis Service

This is a tool that includes several services (plotting, EOF, random forest importance, difference of two fields, lagged correlation). Datasets can be searched by time range, by name, by provider, by variable name, etc. NetCDF files can also be uploaded, and online files served by OPeNDAP can also be used in the analysis.

Select Service: Color Plot Number of datasets: 1

Select Datasets and Variables

Search Filename/varNames/title:

Filter by Variables Short Names:

Filter by Variables Longname:

Select Datasets:

Pick a basin:

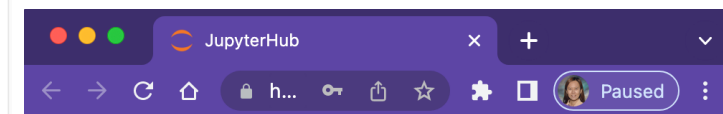
Display Options:

Color Level Type (just an example): 10 levels Modify

Specify Colormap: rainbow

Plot Title:

Analysis Purpose:



Sign in

Username:

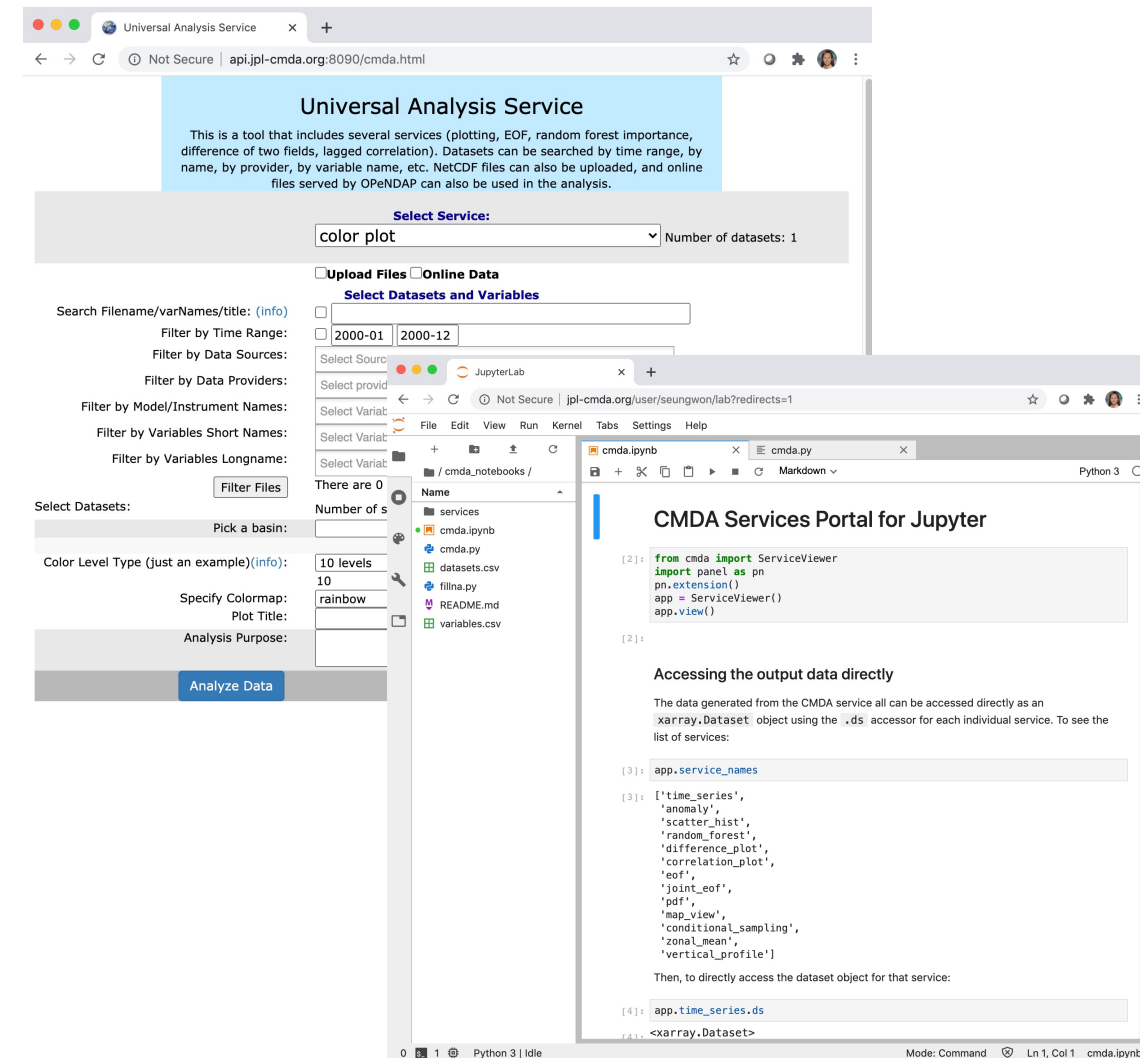
Password:

Group Project Server

- Project Server: <https://jpl-cmda.org>
- Group Project Page: <https://jpl-cmda.org/ccs.html>
- Webservice HTML interface: <https://jpl-cmda.org/cmda.html>
- Jupyter Notebook interface: <https://hub.jpl-cmda.org>

Climate Model Diagnostic Analyzer (CMDA)

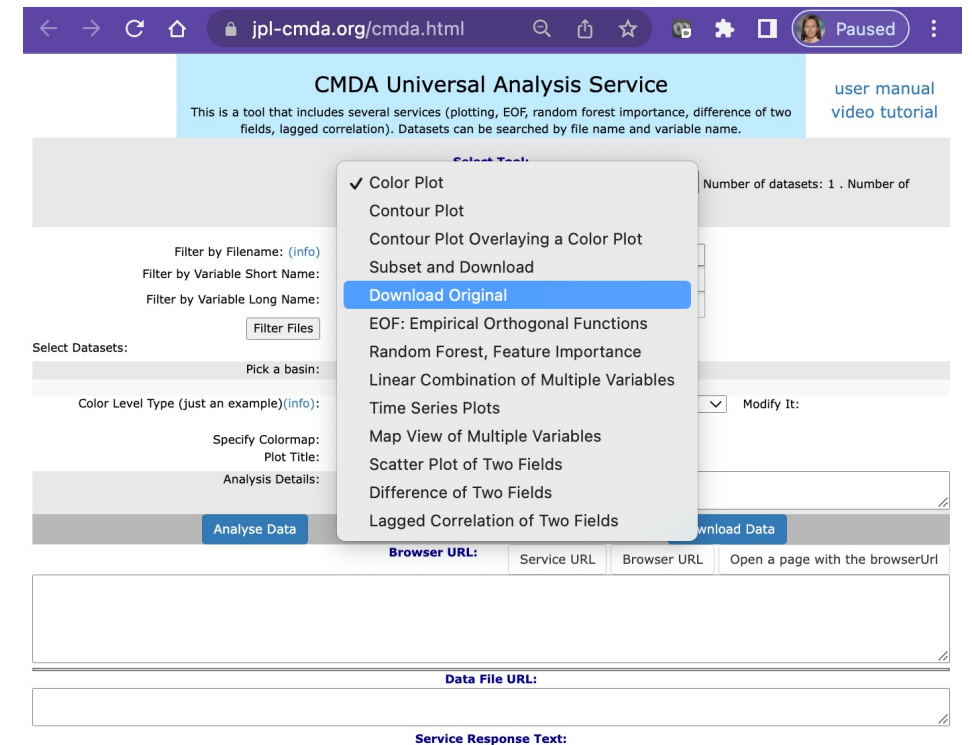
- System designed to analyze climate datasets (monthly, lat-lon-press gridded datasets).
- Server running on Amazon cloud AWS.
- Datasets stored in AWS data storage S3.
- The only requirement from a user machine is a web browser with an internet connection.
- Provides datasets and analysis services.
- You can analyze the datasets using the services.
- You can download/access analyzed output datasets.
- You can download/access original input datasets.
- Provides two interfaces: HTML-web interface, Jupyter-notebook interface



CMDA Analysis Tools*

- Color Plot
- Contour Plot
- Contour Plot Overlaying a Color Plot
- Subset and Download
- Download Original
- Empirical Orthogonal Function
- Random Forest, Feature Importance
- Linear Combination of Multiple Variables
- Time Series Plot
- Map View of Multiple Variables
- Scatter Plot of Two Variables
- Difference of Two Variables
- Lagged Correlation of Two Variables

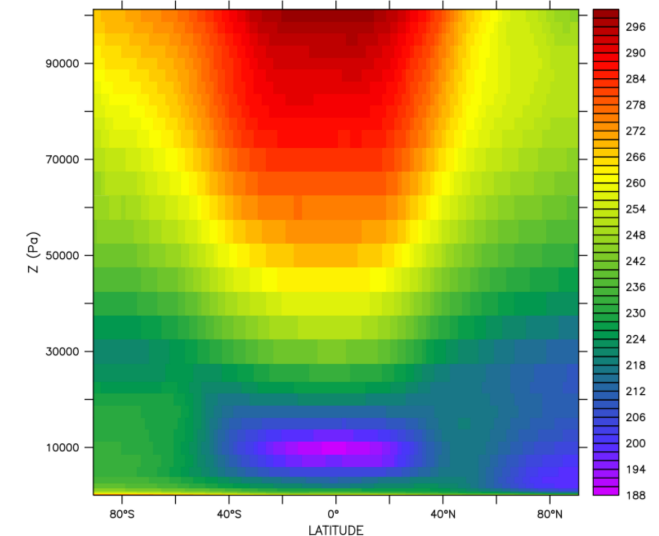
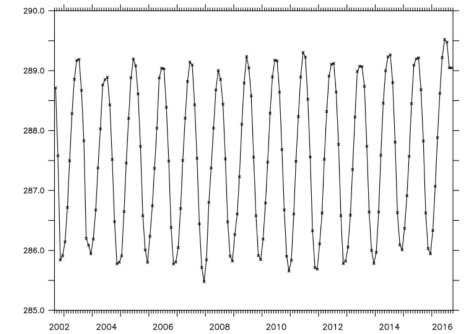
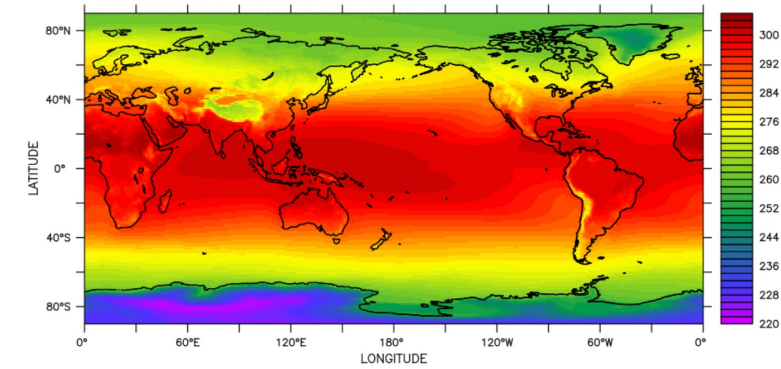
<https://jpl-cmda.org/cmda.html>



*Most tools have subsetting, averaging, and axis-selection options.

CMDA Datasets

- Data Structure: monthly, latitude-longitude-pressure gridded
- Temporal Resolution: monthly
- Temporal Range: various (10 years or longer)
- Spatial Resolution: various (1, 0.5, 0.25 degrees, etc)
- Geographic Range: global
- Dimensions: time, latitude, longitude, pressure
- File Format: NetCDF
- Types: model outputs, reanalysis datasets, observational datasets
- Variables: various ocean, atmosphere, and land variables
- Number of Datasets: 2,145 sets
- Total Data Volume: 2.15 TB
- Typical Data Size: 1 GB per dataset



CMDA Datasets for Summer School

- The datasets used for topic 1,2,3,4,5 are listed in the attached document in the Project Server page.
- https://jpl-cmda.org/2023_datasets_list.pdf
- The document shows Topic Number, Data Name, Data Filename, Data Search Condition in the Universal Analysis Service, and Data Time Range.

A: Topic	B: Dataset (monthly, lat-lon gridded)	C: Data Filenames (under the directory /home/jowyan/cmda_data/cmp5/ in Jupyter Server)	D: Search Conditions in Universal Analysis Webservice	E: Time Range (YYYY-MM-YYYY-MM)
1	CERES TOA incident shortwave radiation	nasa/ceres/nasa_ceres_rsd_200003-201812.nc	ceres, rsdt, 2018	2000-03-2018-12
1	CERES TOA outgoing shortwave radiation	nasa/ceres/nasa_ceres_rslut_200003-201812.nc	ceres, rslut, 2018	2000-03-2018-12
1	CERES TOA outgoing longwave radiation	nasa/ceres/nasa_ceres_rltut_200003-201812.nc	ceres, rltut, 2018	2000-03-2018-12
1	ARGO ocean temperature	others/argo_temperature.nc (then choose ARG0_TEMPERATURE_ANOMALY for variable)	argo	2004-01-2018-12
1	AMSR-E sea surface temperature	nasa/amsre/tos_amsre_l3_v7_200206-201012.nc	amsre	2002-06-2010-12
1	ECMWF sea surface temperature	others/tos_era5.nc	era5	1979-01-2021-04
1	ECMWF surface air temperature	others/tas_era5_1979-2018.nc	era5	1979-01-2018-12
1	NOAA NODC Ocean Heat Content Anomaly	noaa/nodc/ohc700_NOAA_200501-201212.nc	ohc	2005-01-2012-12
1	NOAA NODC Ocean Heat Content Anomaly	noaa/nodc/ohc2000_NOAA_195501-201212.nc	ohc	1955-01-2012-12
2	CERES TOA outgoing clear-sky longwave radiation	nasa/ceres/nasa_ceres_rltucs_200003-201812.nc	ceres, rltucs, 2018	2000-03-2018-12
2	CERES TOA outgoing longwave radiation	nasa/ceres/nasa_ceres_rltut_200003-201812.nc	ceres, rltut, 2018	2000-03-2018-12
2	CERES TOA outgoing clear-sky shortwave radiation	nasa/ceres/nasa_ceres_rslucs_200003-201812.nc	ceres, rslucs, 2018	2000-03-2018-12
2	CERES TOA outgoing shortwave radiation	nasa/ceres/nasa_ceres_rslut_200003-201812.nc	ceres, rslut, 2018	2000-03-2018-12
2	MODIS cloud fraction (liquid + ice)	users/tkubar/output_monthly_modis_terra_l3_cf_day_mean_2000_2019_v6pt1.nc	modis, tkubar, cf, 2019	2000-02-2019-01
2	MODIS liquid cloud fraction	users/tkubar/output_monthly_modis_terra_l3_cf_liquid_mean_2000_2019_v6pt1.nc	modis, tkubar, cf, 2019	2000-02-2019-01
2	MODIS ice cloud fraction	users/tkubar/output_monthly_modis_terra_l3_cf_ice_mean_2000_2019_v6pt1.nc	modis, tkubar, cf, 2019	2000-02-2019-01
2	GPCP precipitation	nasa/gpcp/pr_GPCP-5G_L3_v2.2_*.nc	gpcp	1979-01-2011-06
2	TRMM precipitation	nasa/trmm/pr_TRMM_199801-201803.nc	trmm	1998-01-2018-04
2	GPM precipitation	others/pr_GPM_200006-202101.nc	gpm	2000-06-2021-01
2	AMSR-E sea surface temperature	nasa/amsre/tos_amsre_l3_v7_200206-201012.nc	amsre	2000-06-2010-02
2	ECMWF sea surface temperature	others/tos_era5.nc	era5	1979-01-2021-04
2	CMIP simulations of above variables	many model outputs for different models and variables	cmp5, historical (further filter with different model/variable name)	various
3	GRACE water storage over land	others/nasa_grace_zl_a_200204-202105.nc	grace, zl, 2021	2002-04-2021-05
3	GPCP precipitation	nasa/gpcp/pr_GPCP-5G_L3_v2.2_*.nc	gpcp	1979-01-2011-06
3	TRMM precipitation	nasa/trmm/pr_TRMM_199801-201803.nc	trmm	1998-01-2018-04
3	GPM precipitation	others/pr_GPM_200006-202101.nc	gpm	2000-06-2021-01
3	ECMWF surface air temperature	others/tas_era5_1979-2018.nc	era5	1979-01-2018-12
3	AIRS near-surface air temperature	nasa/airs/tas_AIRS_200209-201609.nc	airs, tas	2002-09-2016-09
3	GLDAS soil moisture	nasa/gldas/ncep_gldas_mrso1_*.nc	gldas, mrso1	1979-01-2019-06
3	SMAP soil moisture	nasa/smap/SMAP_L4_SM_201503-202105_reg_grid.nc	smap	2015-03-2021-05
4	WRF aqua-planet simulations with 8 different physical parameterization	users/ksuselj/ST_4K_v1.nc users/ksuselj/COTie_4K_v1.nc users/ksuselj/COHf_4K_v1.nc users/ksuselj/MPwdm_4K_v1.nc users/ksuselj/MPmor_4K_v1.nc users/ksuselj/PBlnn2_4K_v1.nc users/ksuselj/PBlyu_4K_v1.nc users/ksuselj/Racm_4K_v1.nc	ksuselj	2000-01-2000-12 Year and month do not mean anything in this dataset.
5	GMSL altimetry	GMSL_TPJAOS_S.1_199209-202203.nc	gmsl	1992-09-2022-03
5	GMSL steric sea level	ssh_grids_v1812_199210-201901.nc (Webservice), noaa_gmsl.nc (jupyter)	ssh, 2019	1992-10-2019-01
5	GRACE water storage over ocean	others/nasa_grace_zo_200204-202105.nc	grace, zo, 2021	2002-04-2021-05
5	GRACE water storage over land	others/nasa_grace_zl_a_200204-202105.nc	grace, zl, 2021	2002-04-2021-05

How to select CMDA Datasets

1. Use the Summer School Dataset List Document to see the datasets used in each topic and the search condition value of the dataset (column D).
2. Put the search condition value in “Filter by Filename” in the Universal Analysis Service.
3. Click the “Filter Files”.
4. The list of datasets that meet the search condition will show up below “Select Datasets”.
5. Choose a dataset that you want to use by clicking a box next to it.

The screenshot shows the Universal Analysis Service web interface. The browser address bar displays `jpl-cmda.org/cmda.html`. The page title is "Universal Analysis Service". Below the title, a description states: "This is a tool that includes several services (plotting, EOF, random forest importance, difference of two fields, lagged correlation). Datasets can be searched by time range, by name, by provider, by variable name, etc. NetCDF files can also be uploaded, and online files served by OPeNDAP can also be used in the analysis."

The "Select Service:" dropdown menu is set to "Color Plot". The "Number of datasets:" is 1.

The "Select Datasets and Variables" section includes a search bar with the text "airs" entered. Below the search bar, there are filters for "Filter by Variables Short Names:" and "Filter by Variables Longname:". A button labeled "Filter Files" is highlighted with a red circle.

Below the "Filter Files" button, a table of datasets is displayed. The table has columns: fileName, var name, var long name, units, lon start, lon end, lon res, lat start, lat end, lat res, vertical levels, time start, time end, and var list. The first three rows are highlighted with a green border. The third row is selected, indicated by a blue checkmark in the first column.

	fileName	var name	var long name	units	lon start	lon end	lon res	lat start	lat end	lat res	vertical levels	time start	time end	var list
☐	nasa/airs/hus_AIRS_L3_RetStd-v5_200209-201105.nc	hus	Specific Humidity	1	0.5	359.5	1	0.5	359.5	1	here	20020916	20110516	hus: Specific Humidity
☐	nasa/airs/ta_AIRS_L3_RetStd-v5_200209-201105.nc	ta	Air Temperature	K	0.5	359.5	1	0.5	359.5	1	here	20020916	20110516	ta: Air Temperature
☒	nasa/airs/tas_AIRS_200209-201609.nc	tas	Near-Surface Air Temperature	K	-179.5	179.5	1	-179.5	179.5	1		20020915	20160915	tas: Near-Surface Air Temperature
☐	li/AIRS/SurfAirTemp_DailyMean.nc	sat	Surface Air Temperature (K)		-179.5	179.5	1	-179.5	179.5	1		20020831	20111231	sat: Surface Air Temperature (K)

Below the table, the "Data Subsetting" section includes fields for longitude (deg), latitude (deg), Time (year-month-day), and Preset Time Range. The "Dataset Time Range" is set to "2002-09-15 to 2016-09-15".

The "Display Options:" section includes a dropdown for "Color Level Type (just an example)" set to "10 levels", a dropdown for "Specify Colormap" set to "rainbow", and a field for "Plot Title".

At the bottom, there are buttons for "Analyse Data" and "Download Data".

Jupyter Server Directory

1. /cmda_data:

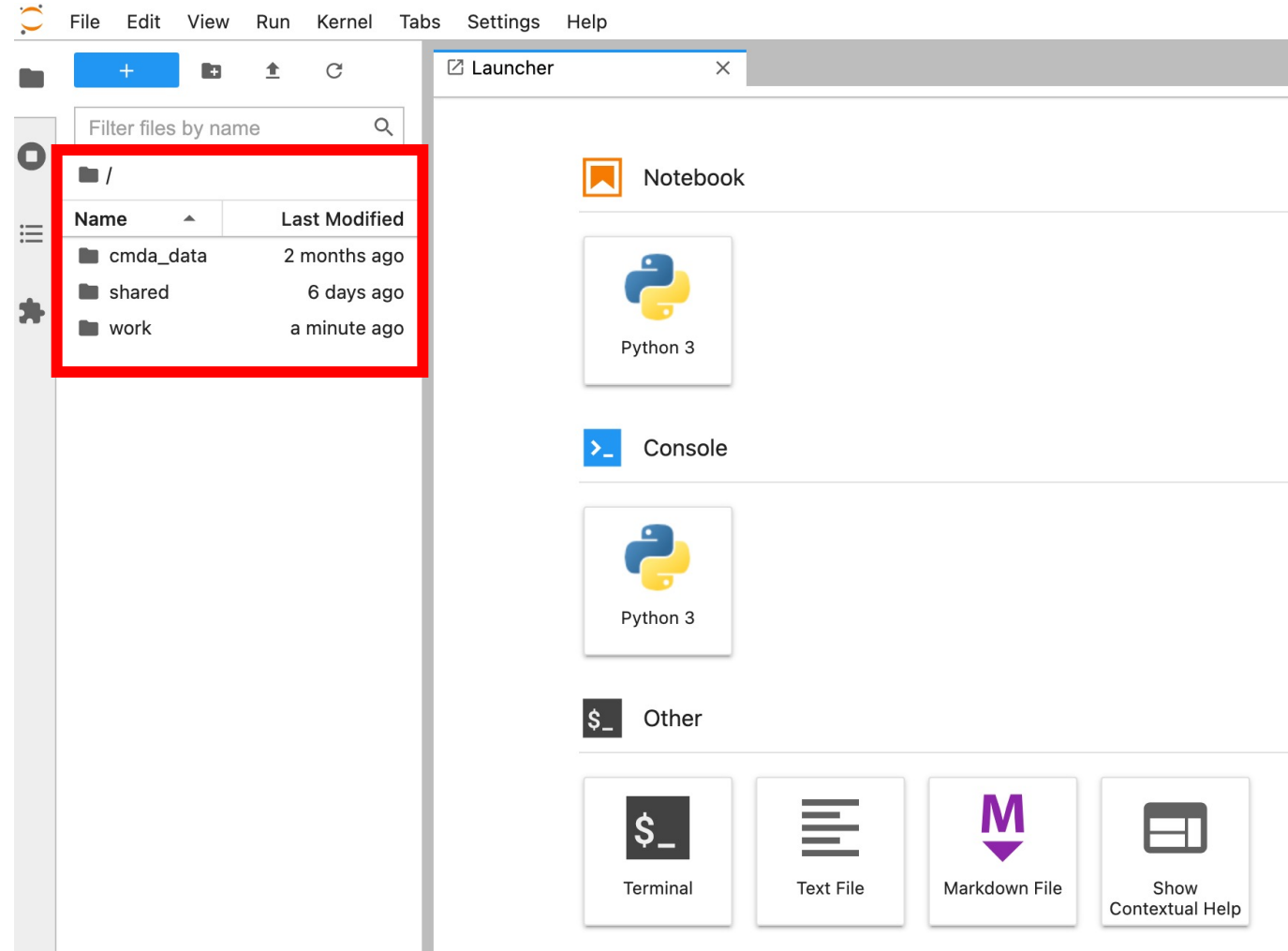
- Contains the subset of CMDA datasets.
- You can directly access them with the home directory path of /home/jovian.
- Look at /work/read_data_example.ipynb for example.

2. /shared:

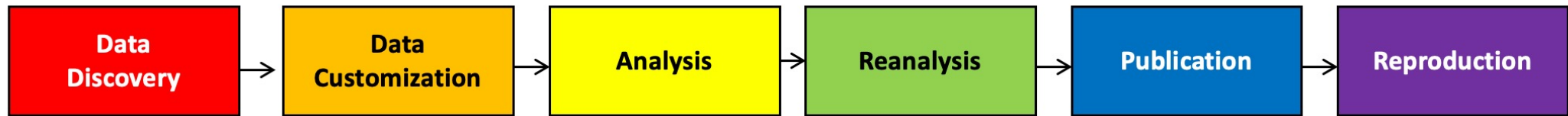
- /shared/NASA_Summer_School_2023:
 - example Jupyter notebooks
- /shared/Groups_2023:
 - shared work directory for each group
- /shared/data:
 - additional data shared for projects

3. /work:

- Your main working directory.
- Contains all the topic example notebooks.
- This is your private directory. Other users cannot access this folder.



Project Approach



- Identify your datasets with the CMDA service (<https://jpl-cmda.org/cmda.html>).
- Explore your datasets with the CMDA service (<https://jpl-cmda.org/cmda.html>).
- Design your own CMDA webservice call (tool selection, input argument selection).
- Execute the webservice call in the CMDA HTML page.
- Write code in Jupyter Notebook, starting with the provided example notebook in /shared/NASA_Summer_School_2023 folder.
- Share your code/results with your group using the Jupyter Server Shared Folder: /shared/Groups_2023/GroupX.
- Discuss the results with your group and prepare a group presentation.

Project Development Team

Please feel free to contact the team if you have any questions.

Seungwon Lee: seungwon.lee@jpl.nasa.gov

(Project in General)

Alex Goodman: alexander.goodman@jpl.nasa.gov

(Jupyter Notebooks, CMDA Webservices)

Group Research Project Topics

- Topic #1: Where is global warming?
- Topic #2: Variability of clouds and precipitation
- Topic #3: Land water storage variability as a function of human and natural controls
- Topic #4: Sensitivity of equilibrium climate on physical parameterizations
- Topic #5: Sea level rise from mass gain and thermal expansion
- Topic #6: Added value of high-resolution downscaling

Group Research Project Scientists

- Topic #1: Dr. Ou Wang (ou.wang@jpl.nasa.gov)
- Topic #2:
 - Dr. Hanii Takahashi (hanii.Takahashi@jpl.nasa.gov),
 - Dr. Jonathan Jiang (jonathan.h.jiang@jpl.nasa.gov)
- Topic #3: Dr. JT Reager (john.reager@jpl.nasa.gov)
- Topic #4: Dr. Kay Suselj (kay.suselj@jpl.nasa.gov)
- Topic #5: Dr. Ben Hemlington (benjamin.d.hamlington@jpl.nasa.gov)
- Topic #6:
 - Dr. Kyo Lee (huikyo.lee@jpl.nasa.gov) ,
 - Alex Goodman (alexander.goodman@jpl.nasa.gov)

Group Project Topic Selection

Topic 1	Topic 2	Topic 3	Topic 4	Topic 5	Topic 6
Global Warming	Clouds	Land Water	Physical Parameterization	Sea-level Rise	High-resolution Downscaling
A link to a Google Document to select your topic will be given in the WebEx chat area during the group formation time.					